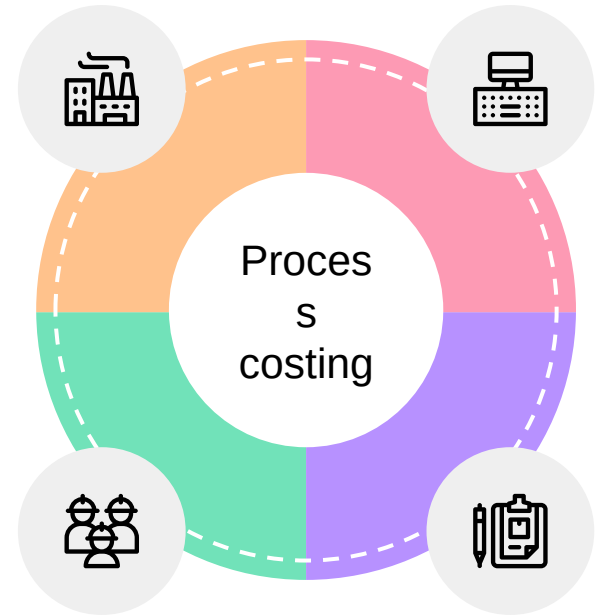


ID interesting case conference

Natchaya Kunanithaworn, MD.

**2nd year pediatric infectious fellow
Division of Infectious Disease, Department of Pediatrics
Faculty of Medicine, Chiang Mai University
27th February 2025**



Identification data

- Case female 8 years old
- Admission date: 24 Dec 2024
- Known case:
- B-cell ALL, SR, 46, XX, last CMT : Maintenance phase week 1 cycle 11 (12/11/67)
- Current medication: TMP-SMX 2 tab po od q Fri-Sun

cc: เจ็บหูซ้าย 2 สัปดาห์ ก่อนมารพ.

Present illness

- **2 wk PTA** (9/12/67) เจ็บหูซ้าย ไปที่ รพช ได้ดูดขี้หู ได้ยาฆ่าเชื้อมาทาน ไม่ทราบชนิด
- **5 day PTA** (19/12/67) เจ็บหูซ้ายที่เดิม มีน้ำไหลออกจากหูซ้ายสีเหลือง เด็กบอกว่าได้ยาลดลง มีบางครั้งที่มีเสียงวิ่งในหู ไม่มีบ้านหมุน มารดาไม่ได้วัดไข้ แต่รู้สึกบางที่ตัวอุ่นๆ ไป รพช. ได้ยา Augmentin(1000) 1x2 po pc
- **2 day PTA** มีแผลในปาก จากเดิมที่ไม่มี ไม่มีไอ/น้ำมูก/ผื่น
- **1 day PTA** อาเจียน 3 รอบเป็นเศษอาหาร ไม่มีปวดท้องไม่มีถ่ายเหลว กินได้น้อยลง กินข้าวได้ 2-3 คำ มีกลิ่นไส้ ยังเจ็บหูซ้ายที่เดิม เจ็บมากขึ้น มีน้ำไหลออกจากหูซ้ายสีเหลือง

Physical Examination at CMU

- **V/S:** BT 36.4 C, BP 108/77 mmHg, PR 107 bpm, RR 20/min, SpO2 100% RA
- Weight 27.3 kg (P 75), Height 127.5 cm (P 50)
- **GA:** normal consciousness
- **HEENT:** mildly pale conjunctiva, no injected pharynx, no icteric sclera, oral ulcer
- **Heart:** regular, no murmur
- **Lung:** Clear and equal breath sound both lungs
- **Abdomen:** Soft, not distended, normal bowel sound
- **Neurological:**
 - Good consciousness, pupil 3 mm RTLBE, no nystagmus
 - no facial palsy, no tongue deviation
 - motor power gr V/V all
 - no ataxia, normal gait
 - no sign of meningeal irritation



Ear exam



Lt ear

- **Left ear:** seen **yellowish clear discharge**, EAC redness and swelling, +/- cloudy TM
- **Right ear:** normal EAC, TM intact

Impression at ward

- Chronic otitis media ? with +/- effusion Lt ear
- Oral ulcer ddx herpes gingivostomatitis

Definition

- chronic otitis media (COM): onset > 3 months
- Acute otitis media: within first 3 weeks

Initial investigations

CBC ...	
☒ Hb	8.1
☒ Hct	24.8
☒ WBC	5130
☒ NRBC per 100 WBC	
☒ Corrected White Blood Cell	
Differential Per 100 WBC ...	
☒ Blast	
☒ Promyelocyte	
☒ Myelocyte	
☒ Metamyelocyte	
☒ Unidentified cell	
☒ Band	
☒ Neutrophil	50.7
☒ Eosinophil	2.5
☒ Basophil	0.2
☒ Lymphocyte	13.1
☒ Monocyte	33.5
☒ Atypical Lymphocyte	
☒ Platelet count	260000
☒ Platelet comment	
☒ Platelet in smear	
☒ RBC	2.46
☒ MCV	100.8
☒ MCH	32.9
☒ MCHC	32.7
☒ RDW	18.5
☒ Abnormal RBC	

Pus Lt EAC:

- G/S: no bacteria found
- C/S: no growth in 2 days

Initial management

- Empirical treatment with piperacillin/ tazobactam 300 MKDAY IV q 8 hr
- Floxy ear drop 5 drop Lt bid
- Acyclovir 30 MKDay IV q 8 hr for 7 days
- Consult ENT

CT temporal bone (including internal acoustic canals) (27/12/67) R/O skull base osteomyelitis



IMPRESSION:

- Suspected **left otomastoiditis with skull base osteomyelitis and associated thrombosis of the left sigmoid sinus.**
- Early **subperiosteal abscess formation/phlegmon** is suspected.
- However, complicated tumor such as **skull base tumor, hematologic malignancy** involving the skull base with superimpose infection cannot be totally excluded.
- Please correlate with the physical examination/intraoperative findings and tissue diagnosis. Imaging follow up after treatment is recommended.

- Permeative **osteolytic lesion involving the mastoid part of the left temporal bone with cortical destruction** and surrounding extraosseous peripheral enhancing soft tissue, overall measuring **2.8*3.2 cm**
- Multiple enhancing enlarge along bilateral jugular regions, up to 1.2*0.7 cm

Consult ID 27/12/67

Impression: Lt mastoiditis with +/- sigmoid sinus thrombosis

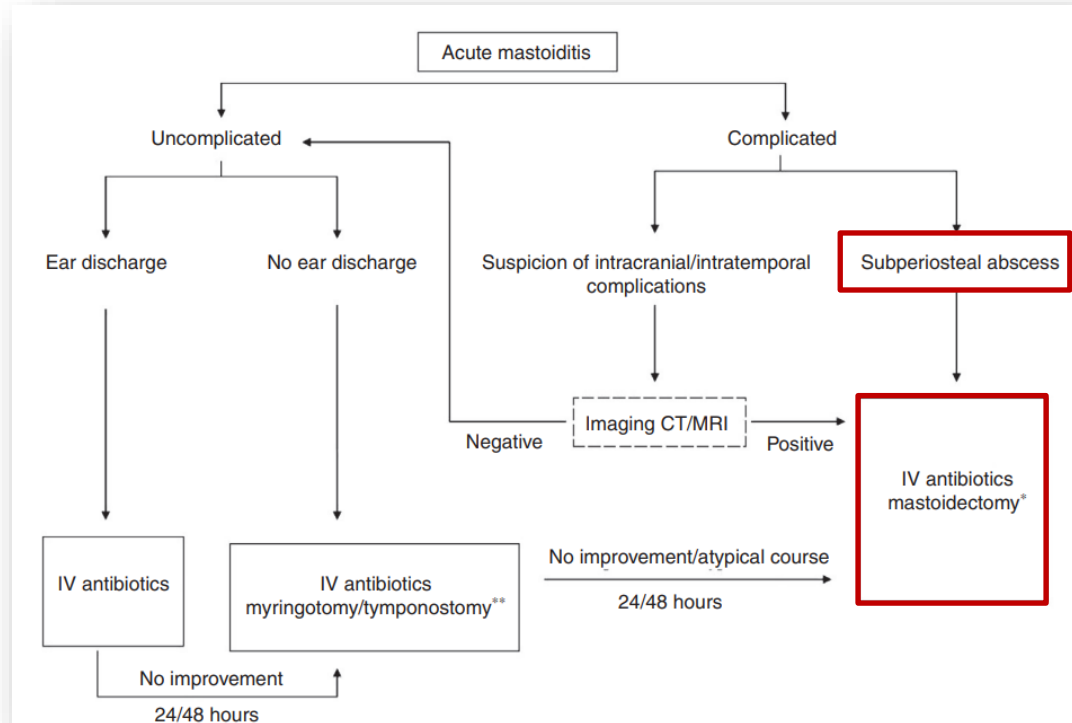
DDx: Acute otomastoiditis in immunocompromised host

- Bacteria: *Pseudomonas aeruginosa*, *S. pneumoniae*, *S. pyogenes*, *H. influenzae*, *M. catarrhalis*
- TB/NTM
- Fungus

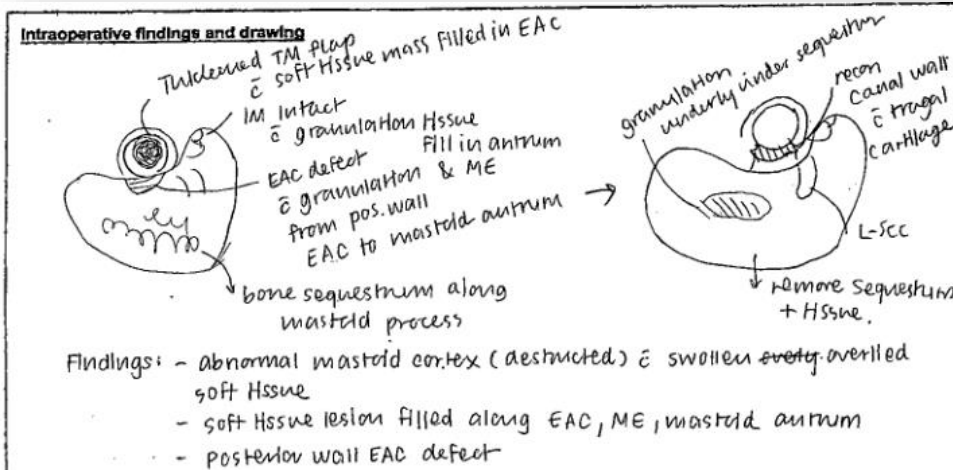
Management:

- Suggest continue Piperacillin/ tazobactam 300 MKDay IV
- If go on surgery → request tissue/pus G/S, culture, wright stain, C/S for fungus, +/- molecular technique

Management algorithm of acute otomastoiditis in children



S/P Lt simple mastoidectomy 28/12/67



Tissue from Lt mastoid

- G/S for bact: no bacteria found, many PMNs
- C/S for bact: NG in 2 days
- Wright stain: not seen fungus
- C/S for fungus: no growth in 2 weeks
- Request tissue for 16srRNA: negative
- Tissue patho

DIAGNOSIS:

1. Tissue from left postauricular region, simple mastoidectomy:

- Keratinous material with inflamed fibrous stroma and exudate, compatible with cholesteatoma
- Negative for malignancy

2. Tissue from left mastoid antrum, simple mastoidectomy:

- Granulation tissue with acute and chronic inflammation
- Negative for malignancy

3. Tissue from left middle ear, simple mastoidectomy:

- Granulation tissue with acute inflammation
- Negative for malignancy

Tissue from Lt mastoid

- G/S for bact: no bacteria found, many PMNs
- C/S for bact: NG in 2 days
- Wright stain: not seen fungus
- C/S for fungus: no growth in 2 weeks
- tissue for 16srRNA: negative
- Tissue patho: granulation tissue with chronic inflammation

S/P Lt simple
mastoidectomy
28/12/67

Fever resolved,
persistent ear
discharge, no pus

6/1/68

Post-op day 11

Fever BT 39.2 C, vomiting,
no otorrhea or D/C

Mx: septic W/U

11/1/68

S/P Explore wound to
DB and biopsy

13/1/68

Persistent fever

Tazocin 300 MKDAY IV q 6 hr

D13

Ciprofloxacin 40 MKDay po bid

D5

Tazocin 300 MKDAY IV q 6 hr

D2

S/P explore wound to debridement and biopsy Lt posterior auricular Lt (13/1/68)

ENT note

- **Findings:** suspected **granulation** at posterior auricular area
- ENT note: intraoperative findings: **suggest atypical infection**
- **DDx**
 - TB
 - Fungus
 - Tumor

Tissue from Lt mastoid

- G/S for bact: no bacteria found, many PMNs
- C/S for bact: NG in 2 days
- Wright stain: not seen fungus
- C/S for fungus: no growth in 2 weeks
- tissue for 16srRNA: negative
- Tissue patho: granulation tissue with chronic inflammation

S/P Lt simple
mastoidectomy
28/12/67

Fever resolved,
persistent ear
discharge, no pus

6/1/68

Fever BT 39.2 C, vomiting,
no otorrhea or D/C
Mx: septic W/U

11/1/68

S/P Explore wound to
DB and biopsy

13/1/68

Persistent high grade
fever, peak BT 39.9 C, no
ear pain, no headache

15/1/68

Tazocin 300 MKDAY IV q 6 hr

D13

Tazocin 300 MKDAY IV q 6 hr

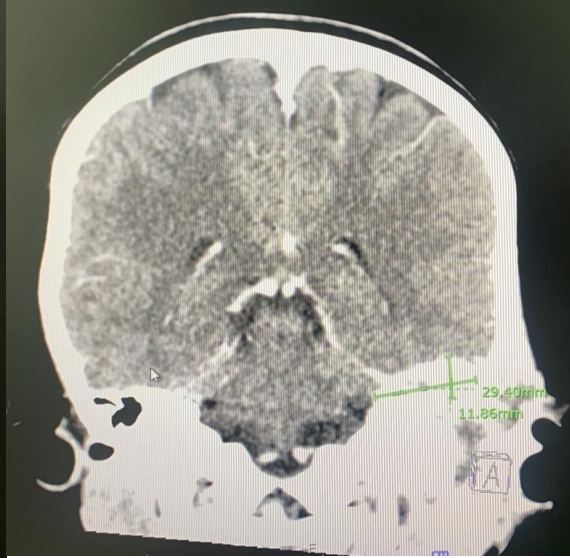
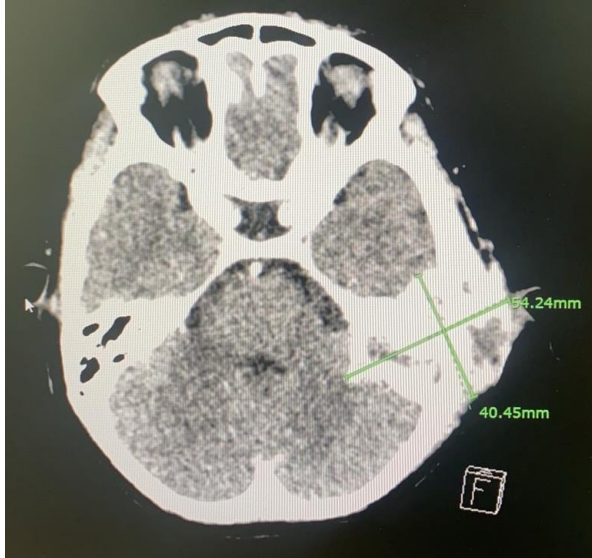
D2

Cipofloxacin 40 MKDay po bid

- Step up to Meropenem 120 MKDay IV q 8 hr
- Request CT brain with contrast

Persistent fever

CT brain with contrast and CTV brain (16/1/68)



Impression:

- Progression of heterogeneous enhancing lesion mainly involving mastoid part of left temporal bone extending to left mastoid sinus and jugular bulb with epidural enhancing soft tissue, prefer to be atypical infection
- Increased size of heterogeneous enhancing lesion involving entire left mastoid air cells extending to mastoid tip, middle ear cavity, measuring $5.4 \times 4.0 \times 5.3$ cm, (previous 2.8×3.2 cm)
- Erosion of left tegmen tympani and mastoid part of left temporal bone → showing epidural heterogeneous enhancing soft tissue along posterior left temporal convexity, measuring $3.6 \times 2.9 \times 1.2$ cm

Re-evaluate

CBC	17/1/68
Hb (g/dL)	10.5
Hct (%)	31.1
WBC (cells/cu.mm)	10,070
Neu (%)	90
Eo (%)	2
Ba (%)	-
Lym (%)	4
Mo (%)	4
Platelet (cells/cu.mm)	235,000
MCV	82.5

- W/U ESR 78 mm/hr, CRP 144 mg/L
- Procalcitonin: 0.98
- H/C for bacteria: NG in 2 days

Tissue from post auricular region Lt ear (13/1/68)

- G/S: few PNMs, no bacteria found
- C/S for bacteria: NG in 2 days
- Wright stain: not seen fungus
- Fungus C/S: NG in 2 weeks
- AFB stain: no AFB found
- Mycobacterium C/S: pending
- PCR for 16srRNA: negative
- Tissue patho (report 17/1/68)
 - granulation tissue, acute suppurative on top chronic granulomatous inflammation
 - Tuberculosis
 - AFB stain: AFB 1+
 - Modified AFB, GMS, G/S: no organisms
- PCR TB/NTM: ?



Further investigations W/U pulmonary TB

- Sputum AFB: no AFB found
- Xpert MTB/RIF ultra: MTB not detected
- Mycobacterium C/S: pending



AFB stain positive
DDx: TB, NTM (rapid, slow grower)
Impression: suspected extrapulmonary mycobacterium

Management → presumptive treatment of TB*

- start IRZE, vitamin B6 (17/1/68), waiting for mycobacterium culture
- W/U pulmonary TB

Tissue from Lt mastoid

- G/S for bact: no bacteria found, many PMNs
- C/S for bact: NG in 2 days
- Wright stain: not seen fungus
- C/S for fungus: no growth in 2 weeks
- tissue for 16srRNA: negative
- Tissue patho: granulation tissue with chronic inflammation

S/P Lt simple
mastoidectomy
28/12/67

Fever resolved,
persistent ear
discharge, no pus

6/1/68

Fever BT 39.2 C, vomiting,
no otorrhea or D/C
Mx: septic W/U

11/1/68

S/P Explore wound to
DB and biopsy

13/1/68

Persistent high grade
fever, peak BT 39.9 C, no
ear pain, no headache

15/1/68

17/1/68

Tazocin 300 MKDAY IV q 6 hr

D13

Tazocin 300 MKDAY IV q 6 hr

D2

Ciprofloxacin 40 MKDay po bid

Meropenem
120 MKDay IV
q 8 hr

Tissue AFB positive 1+
DDx. TB, NTM
Mx
• Start IRZE

Persistent fever



Tissue PCR for TB/NTM:

**NTM detected
(report 22/1/68)**

ITEM	RESULT
Mycobacterium tuberculosis complex(MTB)	NOT DETECTED
Non-tuberculosis mycobacteria(NTM)	DETECTED
Tissue#	
(ear)	
(ear)	

DDx. TB, NTM
Mx: **add drug
cover NTM**

21/1/68

23/1/68

Isoniazid

Rifampicin

Pyrazinamide

Off Ethambutol



Add Azithromycin

Add Amikin IV

Meropenem 120 MKDay IV q 8 hr



Tissue PCR for TB/NTM:

NTM detected
(report 22/1/68)

DDx: TB, NTM

Mx: **add drug
cover NTM**

21/1/68

23/1/68

Isoniazid



Rifampicin



Pyrazinamide



Ethambutol



Re-start Ethambutol



Add Azithromycin



Add Amikin IV



Meropenem 120 MKDay IV q 8 hr



Tissue PCR for TB/NTM:

NTM detected
(report 22/1/68)

DDx. TB, NTM

Mx: **add drug
cover NTM**

21/1/68

23/1/68

Isoniazid



Rifampicin



Pyrazinamide



Ethambutol



Re-start Ethambutol



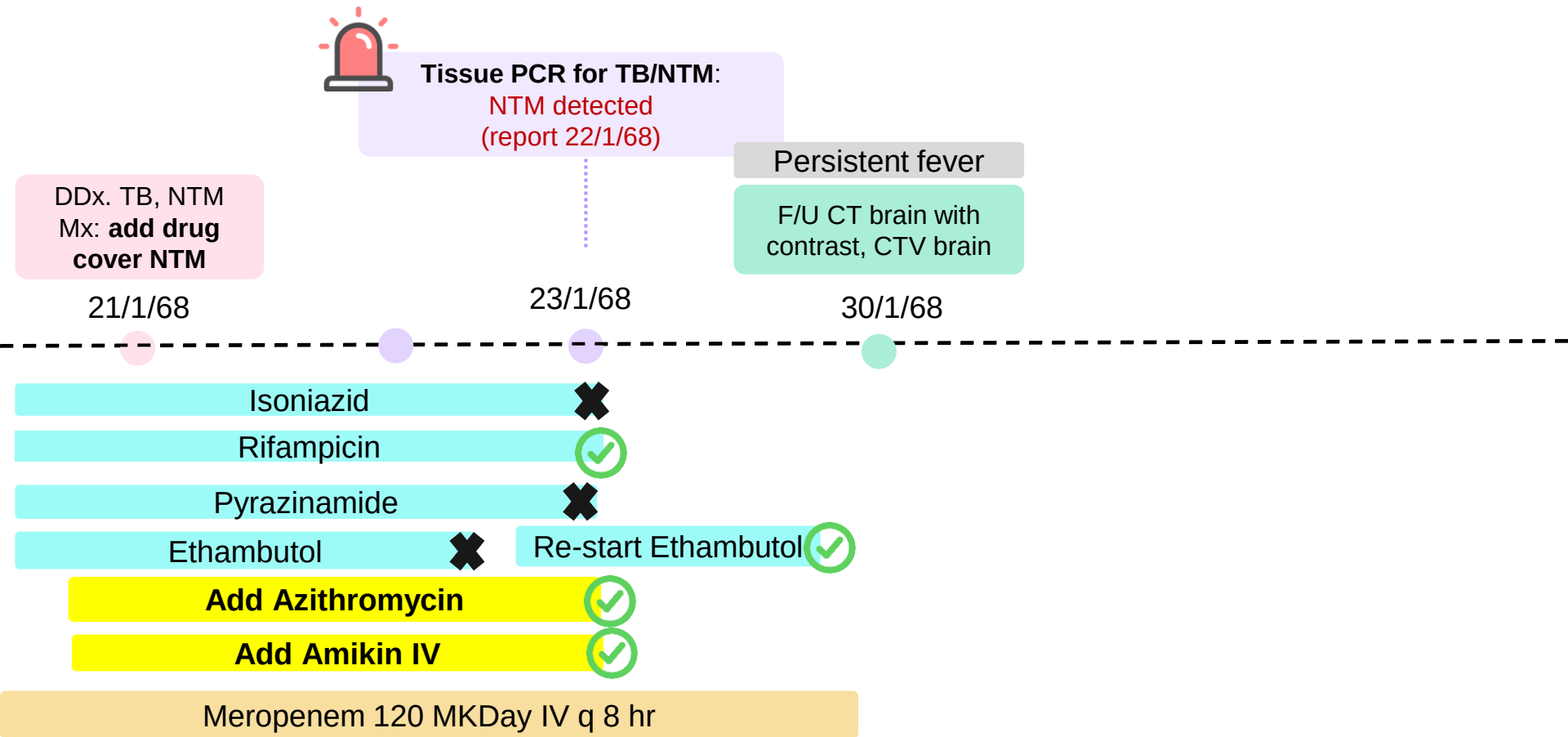
Add Azithromycin



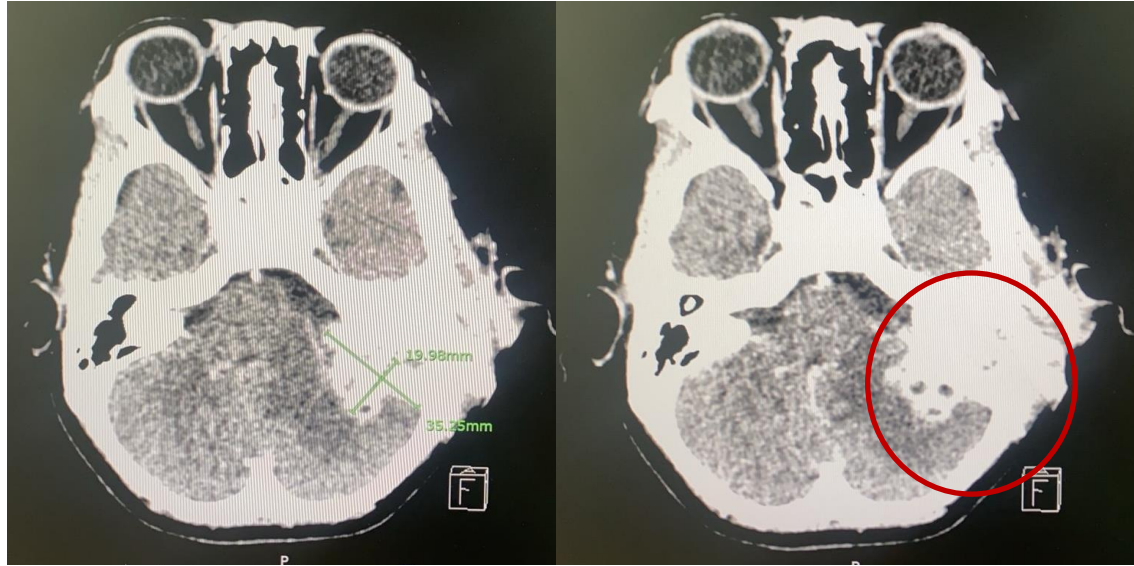
Add Amikin IV



Meropenem 120 MKDay IV q 8 hr



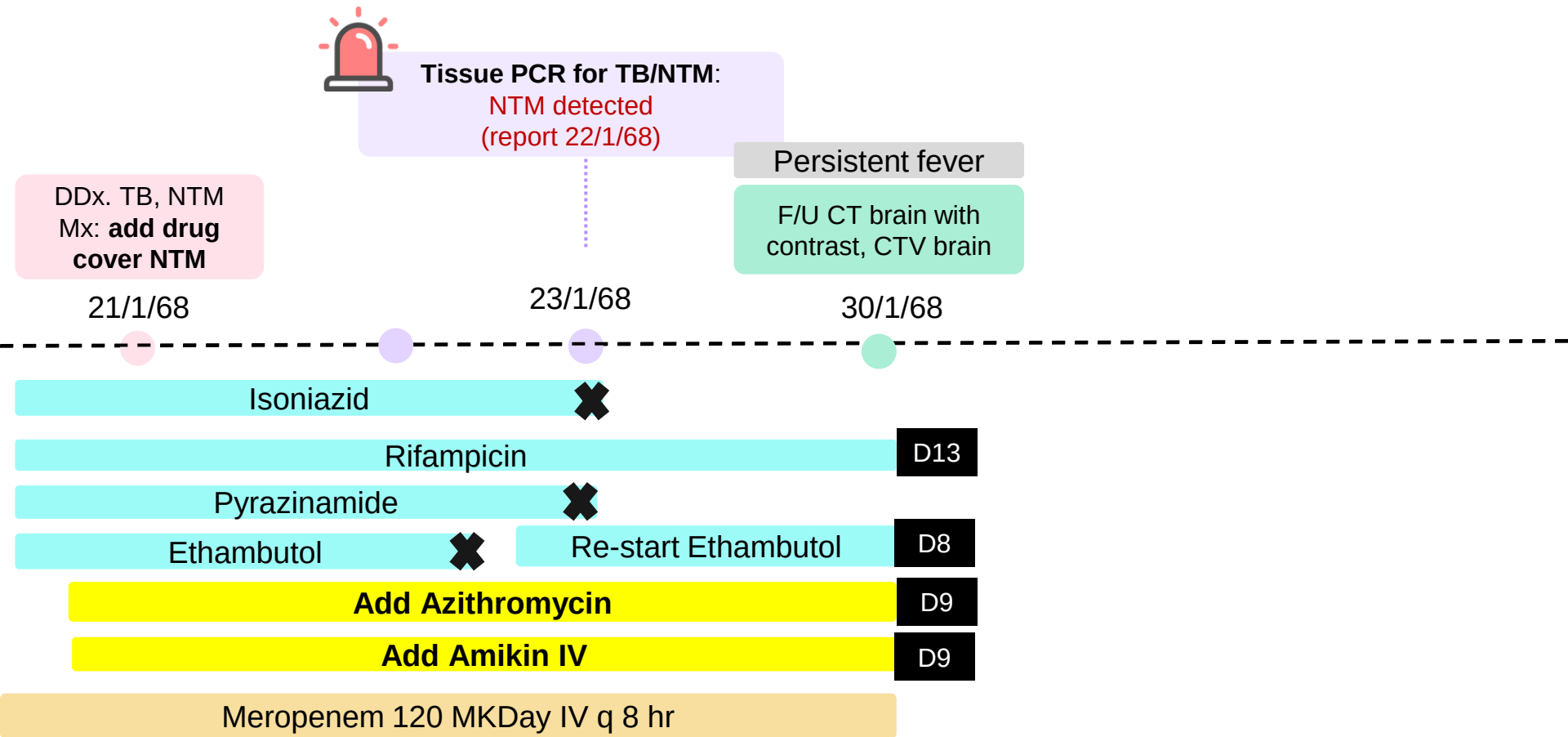
CT brain with contrast + CTV brain (30/1/68)

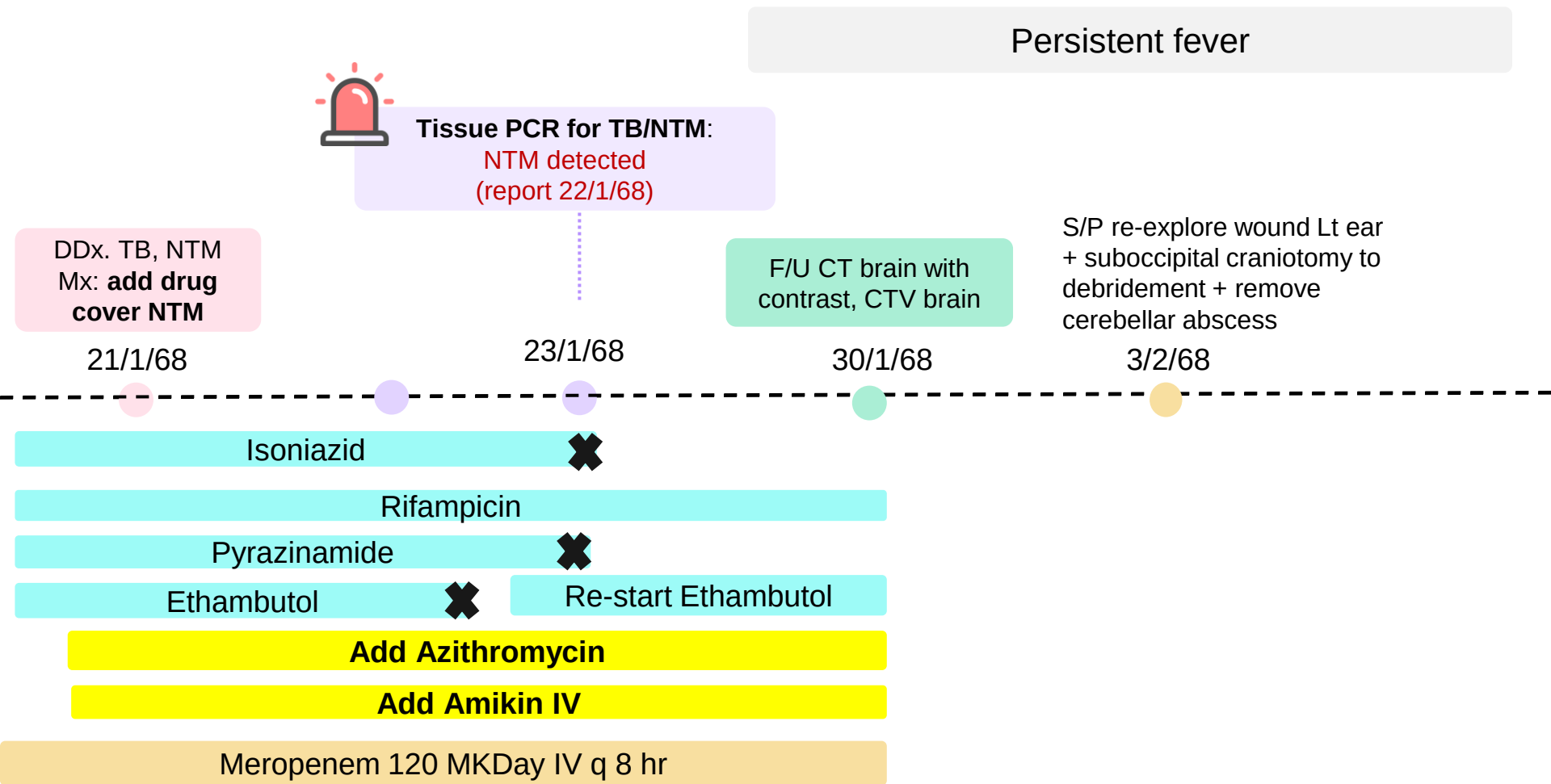


Impression:

- Progression of the pre-existing heterogenous enhancing lesion mainly involving mastoid part of the left temporal bone extending to left sigmoid sinus and jugular bulb with brain parenchyma invasion
- The lesion **prefer to be atypical infection**

- **Increased size** of extension of the **heterogenous lesions involving entire left mastoid air cells** extending to mastoid tip, middle ear cavity
- **Increased size** of **epidural heterogenous enhancing soft tissue** along posterior left temporal convexity, measuring 4.0*3.5*1.4 cm
- Newly seen **multiloculated rim-enhancing hypodense lesions** extending to adjacent brain parenchyma in **left cerebellar hemisphere** associated with brain edema, suspected brain parenchyma invasion





Pus from brain abscess (3/2/68)

- Pus for mycobacterium PCR TB/NTM: **NTM detected** (report 5/2/68)
- Pus for 16srRNA: streptococcus spp.
- Pus C/S for bacteria: NG in 2 days
- Pus for 18srRNA: negative

Tissue from mastoid

- C/S for bacteria: NG in 2 days
- Wright stain: not seen fungus
- Fungus C/S: NG in 2 weeks
- **AFB stain: positive 3+ (report 4/2/68)**
- mAFB: negative
- Mycobacterium C/S: prelim report: **slow growing mycobacteria**

Request

Other# [tissue from mastoid]

Comment

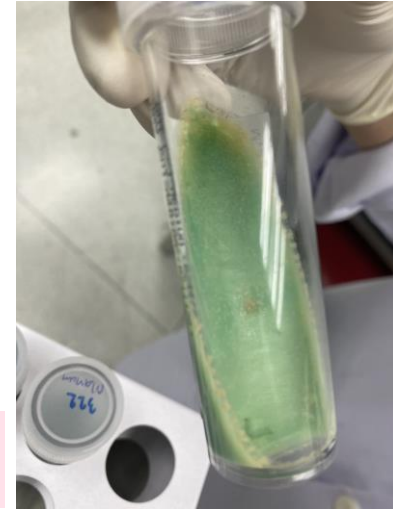
Preliminary report: Slowly Growing Mycobacteria (SGM)

DDx NTM
Slow grower NTM

- MAC
- *M. kansasii*
- *M. marinum*



Tissue mycobacterial culture
(13/1/68): ***M. avium*** report 6/2/68

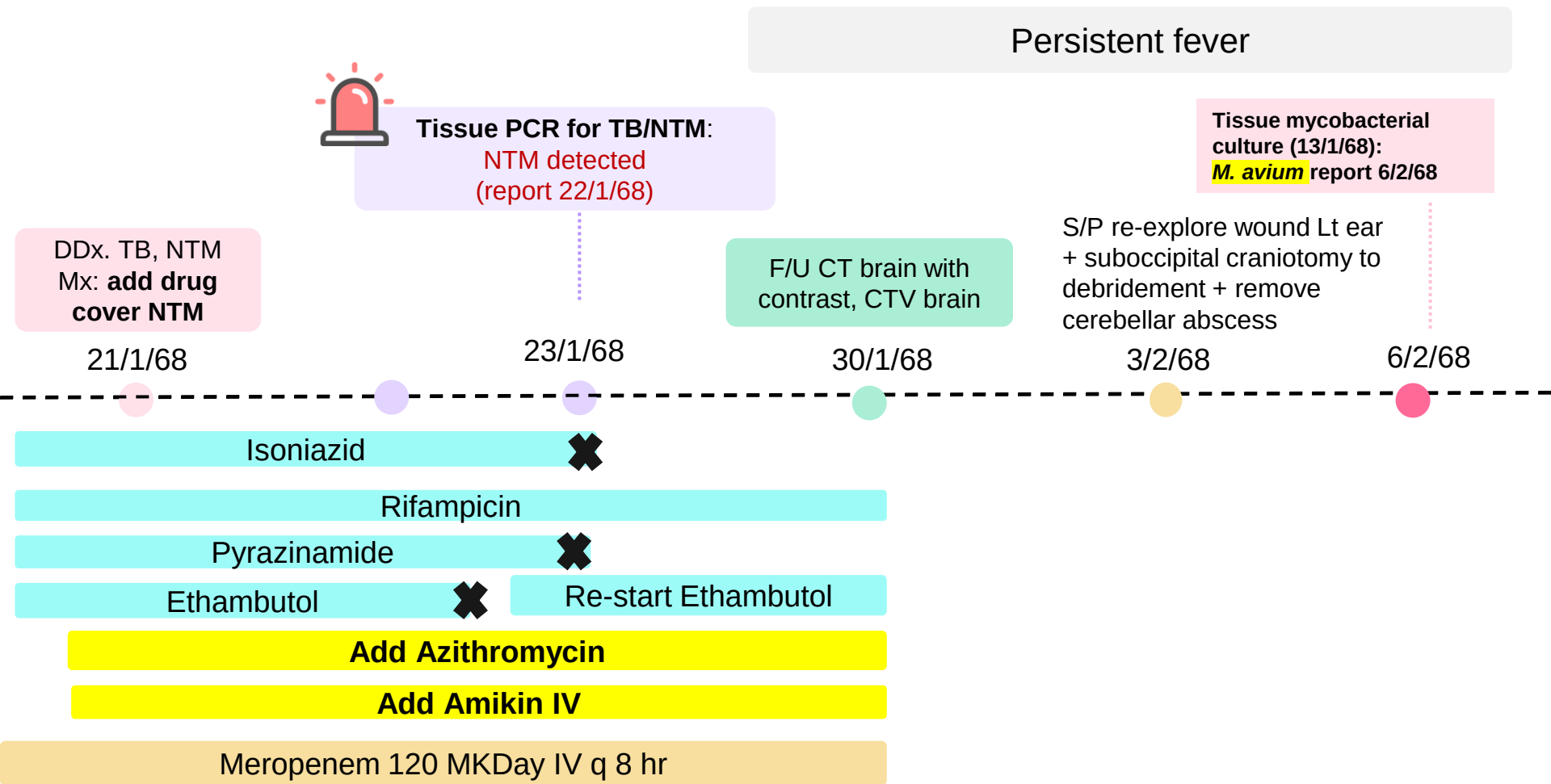


Management

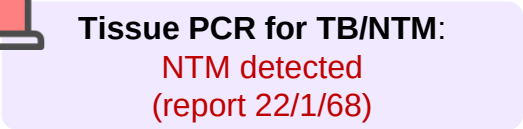
- Impression: NTM otomastoiditis with epidural and Lt cerebellar abscess from *M. avium*
- W/U line probe assay for drug susceptibility
- Treatment for *M. avium*: Macrolide sensitive vs macrolide resistance?

Management in this patient:

- Off amikin
- Add levofloxacin IV
- Continue rifampicin, azithromycin, ethambutol



Persistent fever



Tissue PCR for TB/NTM:
NTM detected
(report 22/1/68)

Tissue mycobacterial culture (13/1/68):
M. avium report 6/2/68

DDx. TB, NTM
Mx: **add drug cover NTM**

F/U CT brain with contrast, CTV brain

S/P re-explore wound Lt ear + suboccipital craniotomy to debridement + remove cerebellar abscess

21/1/68

23/1/68

30/1/68

3/2/68

6/2/68

Isoniazid ✖

Rifampicin 

Pyrazinamide 

Ethambutol 

Ethambutol 

Azithromycin 

Amikin IV D16 

D16 

Meropenem 120 MKDay IV q 8 hr	D24	
-------------------------------	-----	---

D24 

Add Levofloxacin IV 

Progression

- After drainage and add levofloxacin IV total 11 days → fever resolved, no ear pain, no ear discharge
- Continue MAC treatment for 4 drug regimen
 - Switch to oral levofloxacin
 - Continue rifampicin, azithromycin, ethambutol
- D/C 17/2/68
- F/U CT brain with contrast 24/2/68

Progression

Admit for CT brain with contrast (24/2/68)

Clinical:

fever, no ear pain, no headache, no ear D/C

PE: left facial palsy

6/2/68

Tissue mycobacterial
culture (13/1/68)

→ *M. avium* report 6/2/68

20/2/68

Sputum C/S for
mycobacterium (19/1/68)

→ *M. avium* report 20/2/68

Tissue C/S for
mycobacterium (3/2/68)

→ *M. avium* report 20/2/68

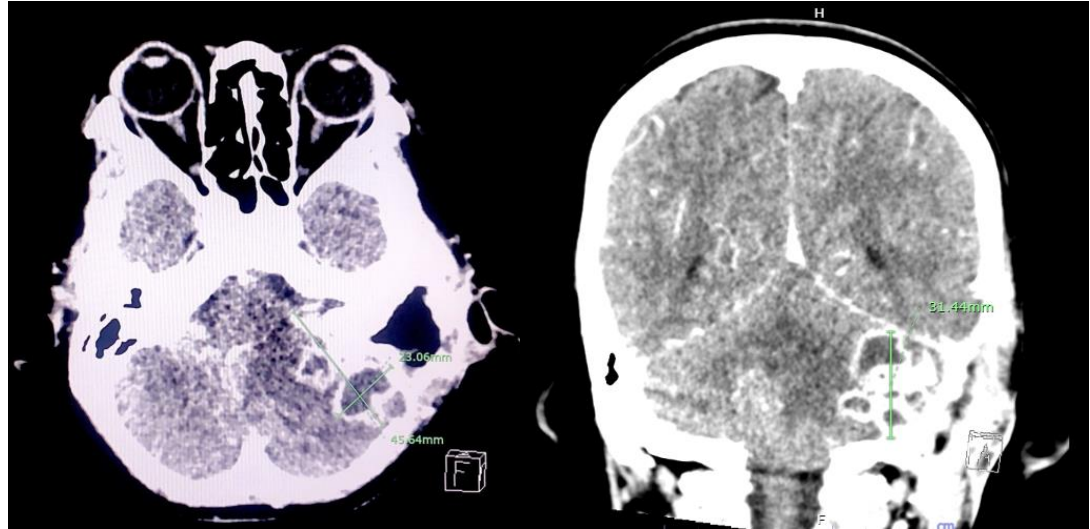
Regimen:

- Rifampicin **D38** (start 17/1/68)
- Azithromycin **D34** (start 21/1/68)
- Ethambutol **D32** (start 23/1/68)
- Levofloxacin IV* 11 days then oral
levofloxacin D7 (total **D18**)

CT brain with contrast (24/2/68)

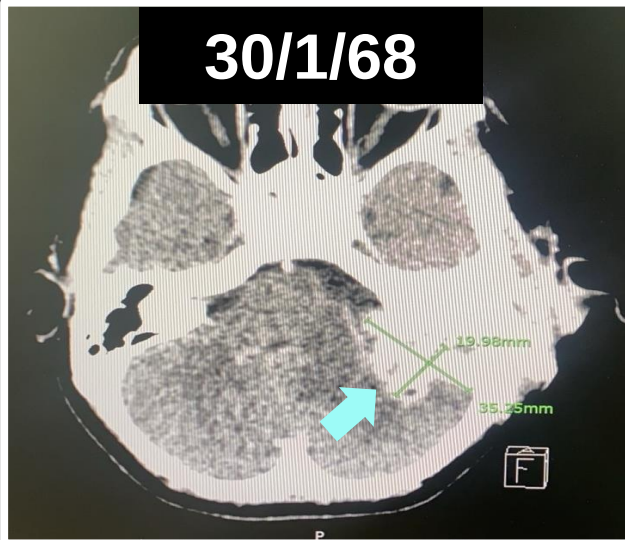


- **Decreased size** of epidural heterogenous enhancing soft tissue along posterior **left temporal convexity**, 1.7*1.6*0.5 cm (previous 4.0*3.5*1.4 cm)
- No extracranial abscess

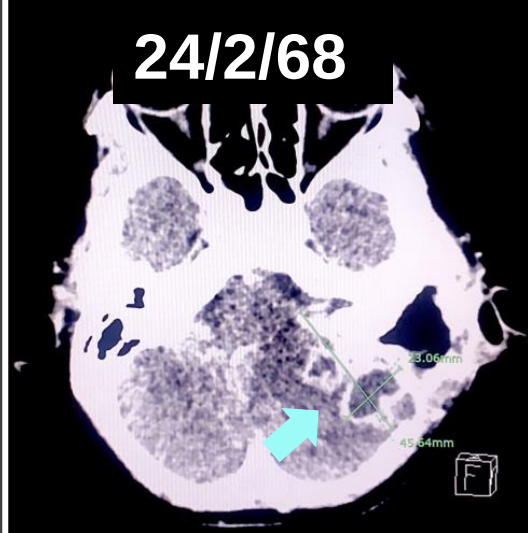


- **Increased size** of multiloculated rim-enhancing hypodense lesions in **the left cerebellar hemisphere** associated with surrounding brain edema, measuring 4.6*2.3*3.1 cm (previous 3.5*2.0*2.5 cm)
- Mass effects: compression and right deviation of the 4th ventricle with **impending left tonsillar herniation**

30/1/68



24/2/68



Plan of management

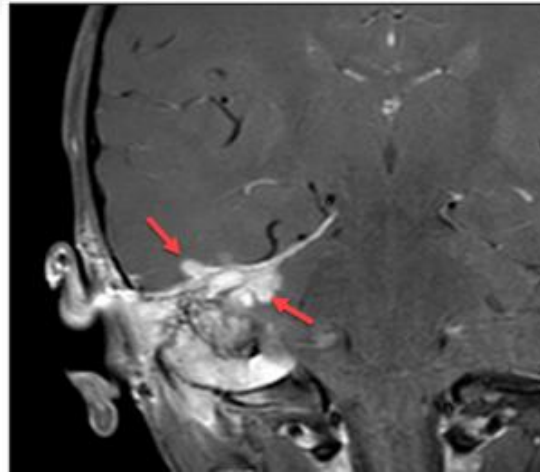
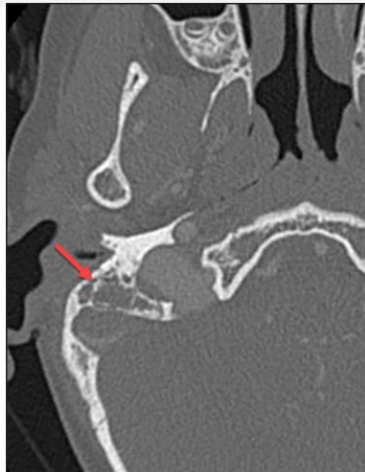
- Waiting for drug susceptibility test (*M. avium*)
- Switch to IV levofloxacin, IV azithromycin
- Continue rifampicin, ethambutol
- F/U CT brain with temporal with contrast next 3 weeks
- Observe neurological signs

Non-Tuberculous Mycobacterial Otomastoiditis with Intracranial Involvement

¹Tushar Chandra, MD; ²Gene A Liu, BS; ³Parth Patel, BS; ⁴Kanika Gupta, MD; ⁵Rahul Nikam, MD

¹Nemours Children's Hospital, Orlando, FL; ^{2,3}University of Central Florida, Orlando, FL; ⁴University of Arizona, Tucson; ⁵Nemours/Alfred I. duPont Hospital for Children, Wilmington, DE

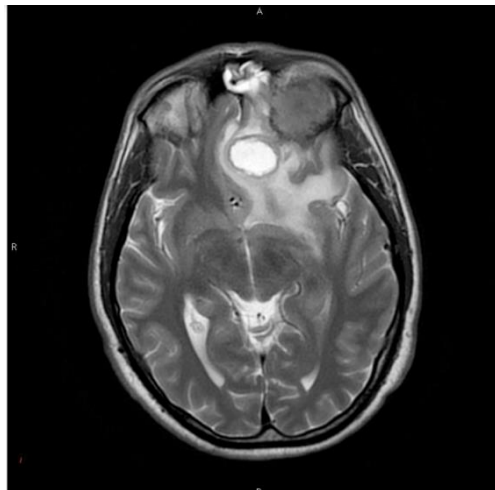
- An 11-year-old presented with right ear pain, intermittent fevers, and new right-sided jaw pain with dizziness.
- *Mycobacterium abscessus*
- **Regimen:** tigecycline, azithromycin, and amikacin





Intracranial abscess due to *Mycobacterium avium* complex in an immunocompetent host: a case report

Mudit Chowdhary^{1*}, Umesh Narsinghani² and Ritu A. Kumar^{1,3}

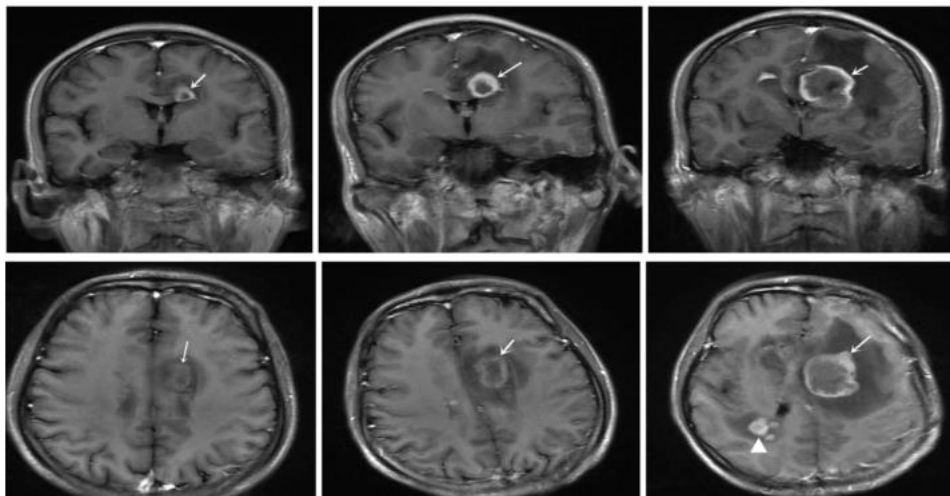


- A 40-year-old Caucasian male presented with a 6-week history of chronic unremitting frontal headaches, followed by left eyelid swelling, blurry vision, and diplopia.
- S/P left frontal craniotomy for partial resection of the masses.
- Tissue AFB positive
- CT brain, CT chest
- Anti-HIV: negative
- *M. avium* complex from brain, sputum
- Regimen: Clarithromycin, Ethambutol, and Rifampin for 12 months
- Outcome: mass effect fully resolved over the following 6- months, sputum AFB negative at 7 months, MRI brain negative at 8 months



Difficulty in diagnosing intracranial infection caused by *Mycobacterium avium* in an AIDS patient: case report and review of the literature

Mengyan Wang¹, Yahui Cui², Jinchuan Shi¹ and Jun Yan^{1*}



- A 31-year-old male, 1st diagnosed HIV, did not start ART
- Presented with throat pain, dysphagia, accompanied by double vision
- Empirical treated toxoplasmosis encephalopathy, fungi, and lymphoma
- the brain tissue culture **detected acid-fast bacilli**.
- Targeted DNA detection identifies it as *M. avium* (*M. avium* subsp. *avium*, *M. avium* subsp. *paratuberculosis*, *M. avium* subsp. *hominissuis*)
- Outcome: died due to cerebral herniation

MAC CNS infection

- Failed to identify an optimal treatment regimen in MAC CNS infections

CDC/ATS/IDSA guideline:

- initiating treatment with at least two drugs for disseminated MAC infection to prevent or delay the emergence of resistance
 - Clarithromycin is the preferred first line agent with Ethambutol as the second recommended drug
 - Adding a third drug, either rifabutin or rifampin, in patients with disseminated MAC and HIV
 - Alternatively, azithromycin can be substituted for clarithromycin as it is efficacious and unlike clarithromycin, does not interact with rifabutin or rifampin in patients with AIDS
-
- The penetration of anti-NTM drugs to CNS infections is further complicated by the relatively impermeable blood brain barrier (BBB).
 - ✓ **Ethambutol and the macrolides only reach sufficient CSF concentrations in the presence of meningeal inflammation**
 - ✓ Although rifampin shows greater efficacy in penetrating the BBB than the other drugs, , it may be below the minimum inhibitory concentration level for some microbial strains

Summary

- NTM otomastoiditis is a rare type of extrapulmonary NTM
- Present with nonspecific symptoms, including **chronic refractory otorrhea, tympanic membrane perforation, ear granulation tissue, and hearing loss**
- A high index of suspicion is needed for diagnosis to avoid the prolonged morbidity it causes.
- Clinical findings of NTM and TB otomastoiditis often overlap, causing diagnostic difficulties.
 - the growth and identification of NTM from the clinical specimens are necessary
 - The culture of mycobacteria is the gold standard for NTM infection diagnoses.
- Surgery (mastoidectomy), which is the mainstay of treatment, should be complemented with antibiotics

Thank you

